Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Select the option that best completes the following sentence: 1 / 1 point

For data with many features, principal components analysis

- ○ identifies which features can be safely discarded
- ○ reduces the number of features without losing any information.
- ○ establishes a minimum number of viable features for use in the analysis.
- ◉ generates new features that are linear combinations of the original features.

Correct
Correct! You can find more information in the lesson on Dimensionality Reduction.

2. Which option correctly lists the steps for implementing PCA in Python? 1 / 1 point

1. Fit PCA to data

2. Scale the data

3. Determine the desired number of components based on total explained variance

4. Define a PCA object

- ◉ 2, 4, 1, 3
- ○ 4, 1, 2, 3
- ○ 2, 1, 3, 4
- ○ 4, 1, 3, 2

Correct
Correct! Note that we need to scale the data prior to fitting a PCA object and obtain the total explained variance afterwards based on the principal components built.

3. Given the following matrix for lengths of singular vectors, how do we rank the vectors in terms of importance? 1 / 1 point

$$\begin{bmatrix} 11 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$v_1, v_2, v_3, v_4$$

- ◉ v_1, v_2, v_3, v_4
- ○ v_4, v_3, v_2, v_1
- ○ v_1, v_4, v_3, v_2
- ○ v_2, v_3, v_4, v_1

Correct
Correct! The bigger the eigenvalue (value on the diagonal), the more important it is.

4. Given two principal components v_1, v_2, let's say that feature f_1 contributed 0.15 to $v_1$and 0.25 to v_2. Feature f_2 contributed -0.11 to v_1 and 0.4 to v_2. 1 / 1 point

Which feature is more important according to their total contribution to the components?

- ○ Neither
- ○ v_1 because $0.15 + 0.25 > -0.11 + 0.4$
- ○ v_2 because $-0.11 + 0.4 < 0.15 + 0.25$

- [x] v_2 because $|-0.11| + |0.4| > |0.15| + |0.25|$

Correct
Correct!

5. (True/False) In PCA, the first principal component represents the most important feature in the dataset.

1 / 1 point

- [x] False
- [ ] True

Correct
Correct! Each principal component in PCA is a linear combination of features in the dataset, so the first one doesn't necessarily correspond to the single most important original feature.